

Pathway Analysis

Juliana Ng
08/17/2026

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Ten Years of Pathway Analysis: Current Approaches and Outstanding Challenges

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What is pathway analysis?

A pathway is a set of genes that work together to carry out a biological function.

Pathway analysis is a bioinformatics analytical approach applied to sequencing data to interpret biological meaning in lists of genes at the gene-set level.

Why?



Reduces complexity



Determines whether a pathway is statistically over- or under-represented in a certain phenotype



Increases explanatory power

Limitations of gene-level statistics

Significance

Not many "significant" genes after multiple hypothesis correction, and biological difference can be small relative to background signal

Interpretability

Long lists of genes without a unifying biological theme

Pathway effect

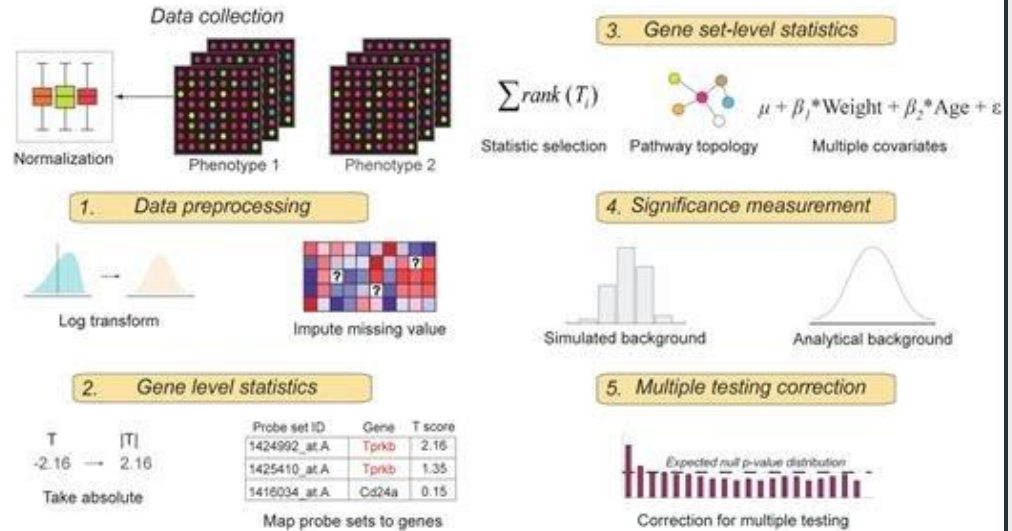
Pathway effect can be missed because genes work together.

Reproducibility

Lists of significant genes are not consistent when the same study is repeated

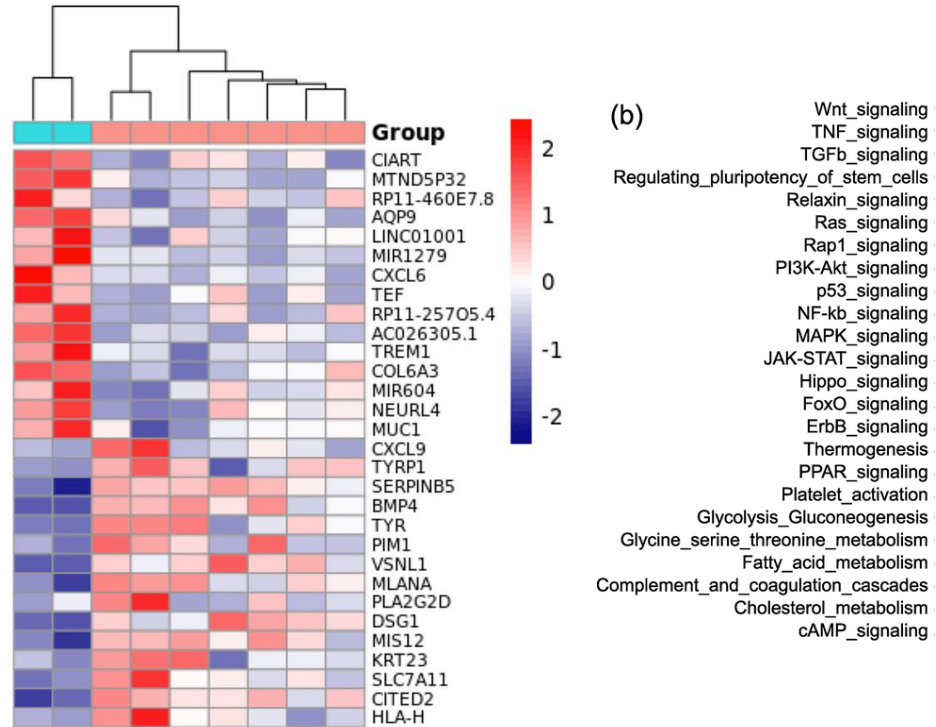
When is it performed?

After differential gene expression analysis.



How?

By correlating gene expression data to pathways in the knowledge base to identify changes in pathways in your data



Three generations of pathway analysis

Over Representation Analysis (ORA)

1st generation 2000-2002

Functional Class Scoring (FCS)

2nd generation 2003-2005

Pathway Topology based approaches

3rd generation 2005

Over Representation Analysis (ORA)

Evaluates a **subset** of differentially expressed genes for over-/under-representation

Tools:

- gProfiler
- Metascape
- clusterProfiler



g:Profiler



Statistics:

- Hypergeometric distribution; Fisher's exact test
- Binomial distribution; Chi-Square test

Most tools are similar because of similar statistics used & overlapping databases

Limitations:

- Genes are treated equally and the statistics consider the number of genes but not the expression values
- Arbitrary threshold > subset of genes > information loss
- Assumes genes act independently
- Assumes pathways act independently

Functional Class Scoring (FCS)

Hypothesis: Large changes have significant effects but weak and coordinated changes in functionally related genes can also have significant effects

Tools:

- GSEA



Statistics:

- Kolmogorov-Smirnov
- Wilcoxon rank sum
- Sum, mean, median, maxmean

Advantages:

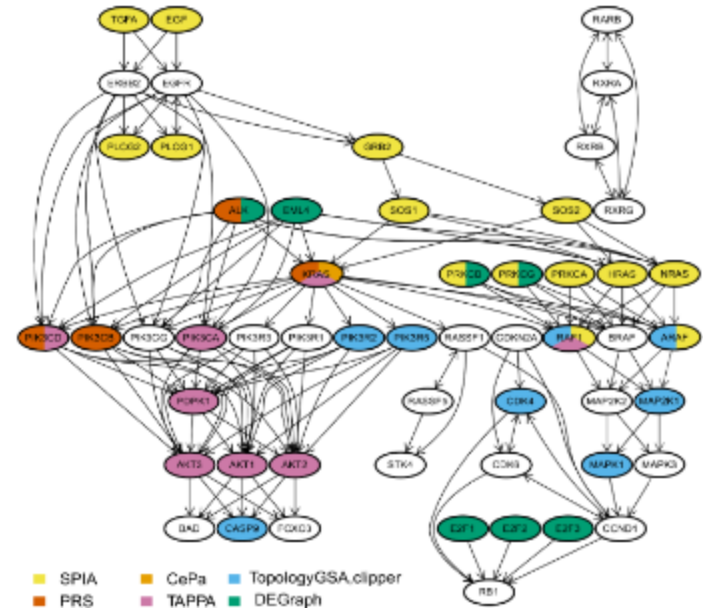
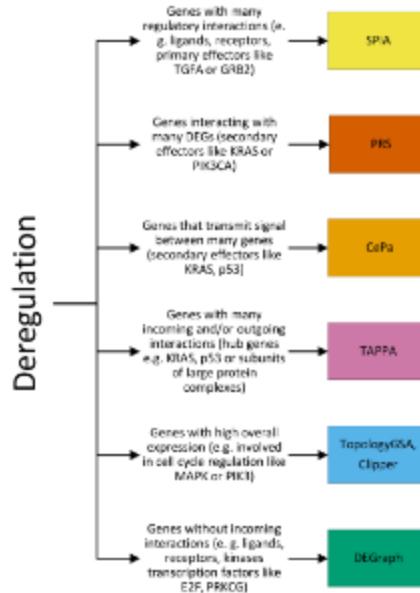
- No arbitrary threshold
- Considers expression values accounting for gene dependence in a pathway

Limitations:

- Assumes pathways act independently
- Genes in different pathways with the same rank are treated equally even if the expression values are different

Pathway topology-based approaches

Computes pathway statistics using network graphs accounting for interactions and directionality of pathways



02

Over Representation Analysis

Some ORA tools

gProfiler

Web-based

Metascape

Web-based

clusterProfiler

R

gProfiler

g:Profiler

[News](#)

[Archives](#)

[Beta](#)

[API](#)

[R client](#)

[FAQ](#)

[Docs](#)

[Contact](#)

[Cite g:Profiler](#)



g:GOST

Functional profiling

g:Convert

Gene ID conversion

g:Orth

Orthology search

g:SNPense

SNP id to gene name

Query

Upload query

Upload bed file

Input is whitespace-separated list of genes ?

```
ENSG00000131055
ENSG00000111596
ENSG00000112062
ENSG00000198400
ENSG00000198719
ENSG00000169174
ENSG00000169071
ENSG00000101421
ENSG00000047936
ENSG00000213398
ENSG00000150712
ENSG00000135049
ENSG00000204580
ENSG00000123384
ENSG00000154153
ENSG00000233333
```

Run query

random example

mixed query example

Options

Organism: ?

Homo sapiens (Human)

☒ Highlight driver terms in GO ?

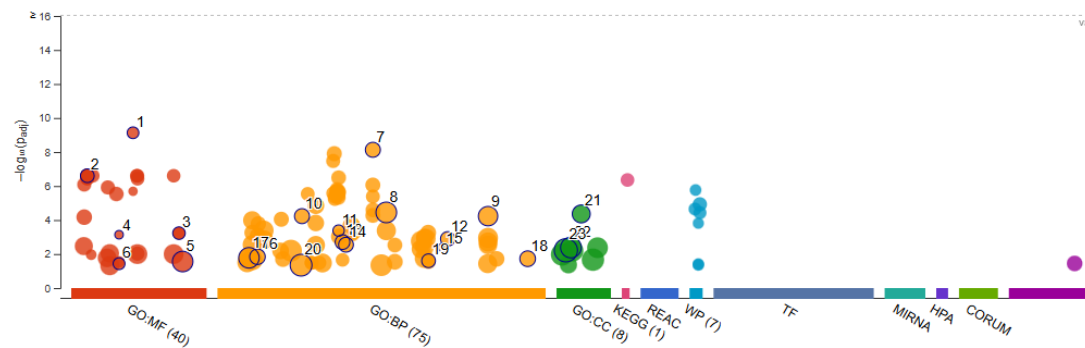
☐ Ordered query ?

☐ Run as multiquery ?

Advanced options ▾

Data sources ▾

Bring your data (Custom GMT) ▾



ID	Source	Term ID	Term Name	Padj (query_1)
1	GO:MF	GO:0034185	apolipoprotein binding	7.401×10^{-10}
2	GO:MF	GO:0005008	hepatocyte growth factor receptor activity	2.486×10^{-7}
3	GO:MF	GO:0071813	lipoprotein particle binding	5.907×10^{-4}
4	GO:MF	GO:0030226	apolipoprotein receptor activity	7.300×10^{-4}
5	GO:MF	GO:0097367	carbohydrate derivative binding	2.753×10^{-2}
6	GO:MF	GO:0030228	lipoprotein particle receptor activity	3.692×10^{-2}
7	GO:BP	GO:0048010	vascular endothelial growth factor receptor signal...	7.390×10^{-9}
8	GO:BP	GO:0051128	regulation of cellular component organization	3.566×10^{-5}
9	GO:BP	GO:1901701	cellular response to oxygen-containing compound	5.890×10^{-5}
10	GO:BP	GO:0030301	cholesterol transport	5.919×10^{-5}
11	GO:BP	GO:0038083	peptidyl-tyrosine autophosphorylation	4.190×10^{-4}
12	GO:BP	GO:0097006	regulation of plasma lipoprotein particle levels	1.281×10^{-3}
13	GO:BP	GO:0042157	lipoprotein metabolic process	1.988×10^{-3}
14	GO:BP	GO:0042632	cholesterol homeostasis	2.829×10^{-3}
15	GO:BP	GO:0090107	regulation of high-density lipoprotein particle ass...	4.048×10^{-3}
16	GO:BP	GO:0008203	cholesterol metabolic process	1.441×10^{-2}
17	GO:BP	GO:0006796	phosphate-containing compound metabolic proce...	1.666×10^{-2}
18	GO:BP	GO:1905952	regulation of lipid localization	1.882×10^{-2}
19	GO:BP	GO:0071827	plasma lipoprotein particle organization	2.486×10^{-2}
20	GO:BP	GO:0030154	cell differentiation	4.511×10^{-2}
21	GO:CC	GO:0043235	receptor complex	4.374×10^{-5}
22	GO:CC	GO:0031410	cytoplasmic vesicle	4.119×10^{-3}
23	GO:CC	GO:0016020	membrane	6.007×10^{-3}

Metascape



Step 1

Upload File Format

Single List:

[.xls/xlsx](#) [.csv](#) [.txt](#)

Multiple List:

[.xls/xlsx](#) [.csv](#) [.txt](#)

Test Upload

[single list](#)

[3 gene lists](#)

Test Identifiers

Gene Symbol [try it!](#)

RefSeq

Entrez Gene ID

Or paste a gene list

ENSG00000073060

ENSG00000165029

ENSG00000145287

ENSG00000102312

ENSG00000062524

Submit

Cancel

Your id type: Ensembl Gene .

Step 2

Optional if you only consider human species in your study.

Input as species:

Any Species

Analysis as species:

H. sapiens (20)

Step 3

Express Analysis

Custom Analysis

Batch Analysis?

Please click the button below to see reports

Analysis Report Page

100

[Gene List Report Excel Sheets](#)[Gene List Report PPT file](#)[All in One Zip File](#)

Warning: this report page can be accessed within the next 72 hours; all data ([session id: t5stpk7qk](#)) will be deleted afterwards.

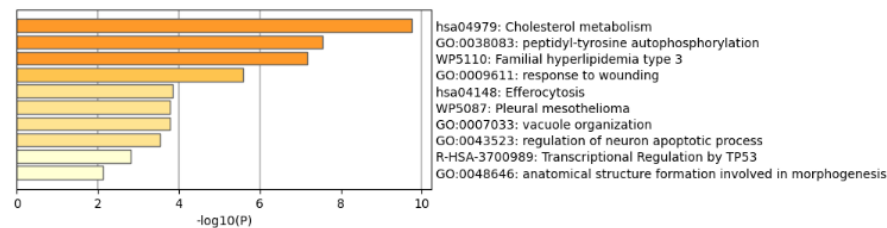
Due to database update, bug fix, and continuous features improvement, Metascape does not provide a mechanism to reproduce the exact same analysis results for the same input data. It is important to save the .zip file as your complete record.

Metascape Gene List Analysis Report

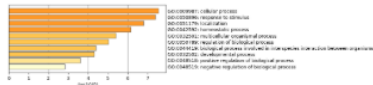
metascape.org¹

Bar Graph Summary

Figure 1. Bar graph of enriched terms across input gene lists, colored by p-values.



The top-level Gene Ontology biological processes can be viewed here.



Gene Lists

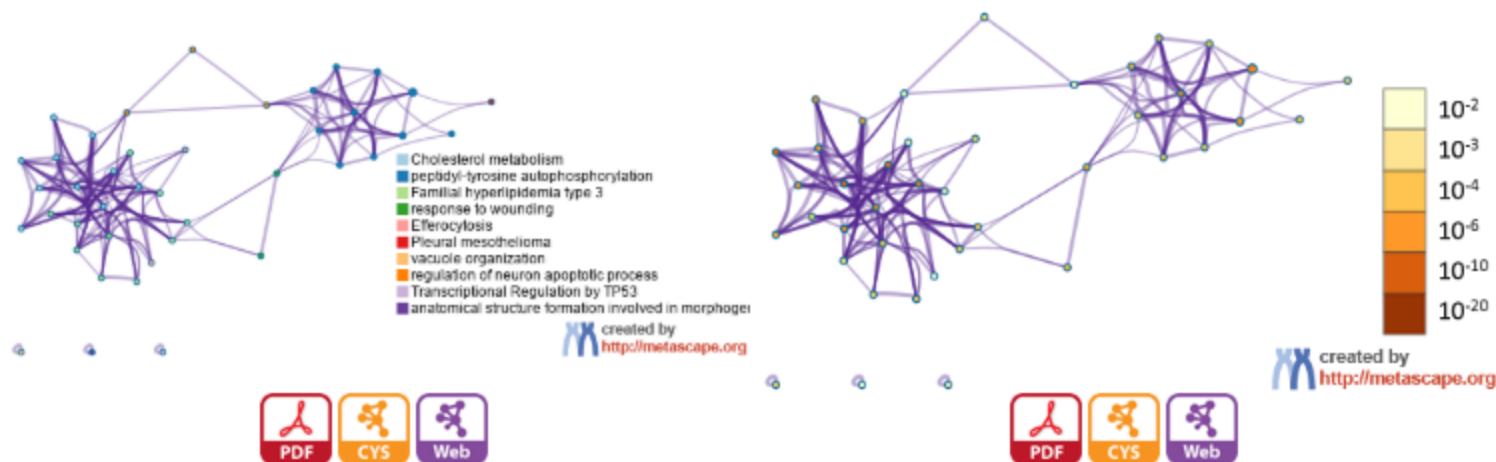
User-provided gene identifiers are first converted into their corresponding H. sapiens Entrez gene IDs using the latest version of the database (last updated on 2026-07-01). If multiple identifiers correspond to the same Entrez gene ID, they will be considered as a single Entrez gene ID in downstream analyses. The gene lists are summarized in Table 1.

Table 1. Statistics of input gene lists.

Name	Total	Unique
MyList	20	20

To further capture the relationships between the terms, a subset of enriched terms has been selected and rendered as a network plot, where terms with a similarity > 0.3 are connected by edges. We select the terms with the best p-values from each of the 20 clusters, with the constraint that there are no more than 15 terms per cluster and no more than 250 terms in total. The network is visualized using Cytoscape⁵, where each node represents an enriched term and is colored first by its cluster ID (Figure 2.a) and then by its p-value (Figure 2.b). These networks can be interactively viewed in Cytoscape through the .cys files (contained in the Zip package, which also contains a publication-quality version as a PDF) or within a browser by clicking on the web icon. For clarity, term labels are only shown for one term per cluster, so it is recommended to use Cytoscape or a browser to visualize the network in order to inspect all node labels. We can also export the network into a PDF file within Cytoscape, and then edit the labels using Adobe Illustrator for publication purposes. To switch off all labels, delete the "Label" mapping under the "Style" tab within Cytoscape, and then export the network view.

Figure 2. Network of enriched terms: (a) colored by cluster ID, where nodes that share the same cluster ID are typically close to each other; (b) colored by p-value, where terms containing more genes tend to have a more significant p-value.



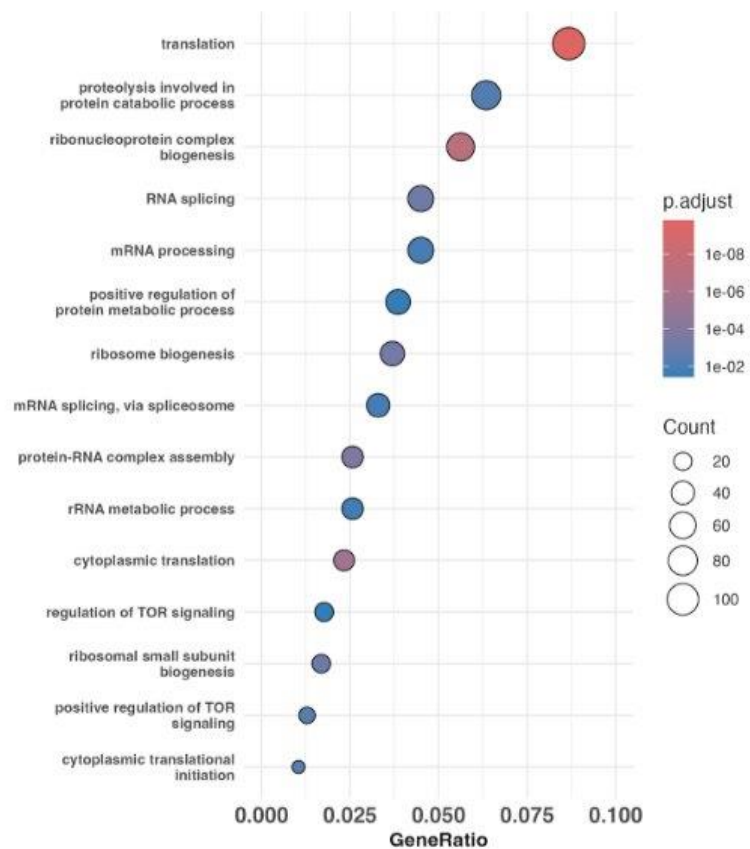
clusterProfiler

```
enrichGO(  
  gene,  
  OrgDb,  
  keyType = "ENTREZID",  
  ont = "MF",  
  pvalueCutoff = 0.05,  
  pAdjustMethod = "BH",  
  universe,  
  qvalueCutoff = 0.2,  
  minGSSize = 10,  
  maxGSSize = 500,  
  readable = FALSE,  
  pool = FALSE  
)
```

Arguments

gene	a vector of entrez gene id.
OrgDb	OrgDb
keytype	keytype of input gene
ont	One of "MF", "BP", and "CC" subontologies.
pvalueCutoff	Cutoff value of pvalue.
pAdjustMethod	one of "holm", "hochberg", "hommel", "bonferroni", "BH", "BY", "fdr", "none"
universe	background genes
qvalueCutoff	qvalue cutoff
minGSSize	minimal size of genes annotated by Ontology term for testing.
maxGSSize	maximal size of genes annotated for testing
readable	whether mapping gene ID to gene Name

	A	B	C	D	E	F	G	H	I	J	K	L
1	ID	Description	GeneRatio	BgRatio	RichFactor	FoldEnrichment	zScore	pvalue	p.adjust	qvalue	geneID	Count
2	GO:0050863	regulation of T cell activation	25/122	404/18860	0.061881	9.566223016	14.04394	8.82E-18	2.28E-14	1.63E-14	HLA-DQA1	25
3	GO:0022409	positive regulation of cell-cell adhesion	22/122	343/18860	0.06414	9.915404101	13.44561	4.79E-16	3.83E-13	2.74E-13	HLA-DQA1	22
4	GO:0051251	positive regulation of lymphocyte activation	22/122	346/18860	0.063584	9.829432389	13.37514	5.75E-16	3.83E-13	2.74E-13	HLA-DQA1	22
5	GO:0007159	leukocyte cell-cell adhesion	24/122	439/18860	0.05467	8.451398484	12.7465	7.10E-16	3.83E-13	2.74E-13	HLA-DQA1	24
6	GO:0002443	leukocyte mediated immunity	25/122	488/18860	0.05123	7.91957807	12.49652	7.41E-16	3.83E-13	2.74E-13	NCR1/TLR	25



03

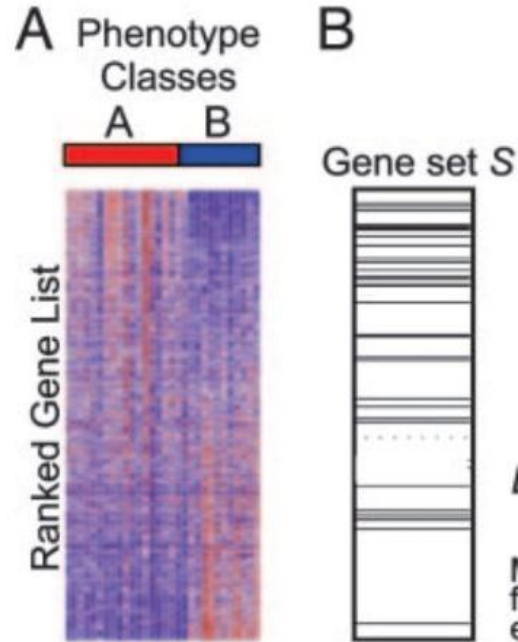
Gene set enrichment analysis: A knowledge-based approach for interpreting genome-wide expression profiles

Aravind Subramanian^{a,b}, Pablo Tamayo^{a,b}, Vamsi K. Mootha^{a,c}, Sayan Mukherjee^d, Benjamin L. Ebert^{a,e}, Michael A. Gillette^{a,f}, Amanda Paulovich^g, Scott L. Pomeroy^h, Todd R. Golub^{a,e}, Eric S. Lander^{a,c,i,j,k}, and Jill P. Mesirov^{a,k}

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Contributed by Eric S. Lander, August 2, 2005

"The goal of GSEA is to determine whether members of a gene set S tend to occur toward the top (or bottom) of the list L, in which case the gene set is correlated with the phenotypic class distinction."

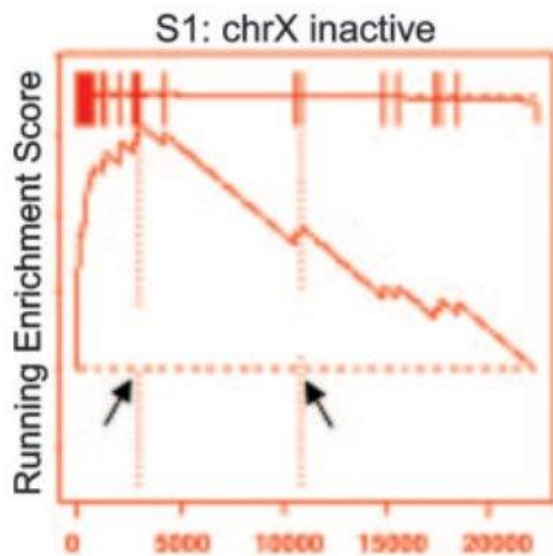


The image shows a computer monitor with a light gray bezel and a matching stand. On the screen, three identical tables are displayed side-by-side. Each table has an orange border and a header row with two columns: 'Gene symbol' and 'Ranking'. The tables are empty, with only the header rows visible.

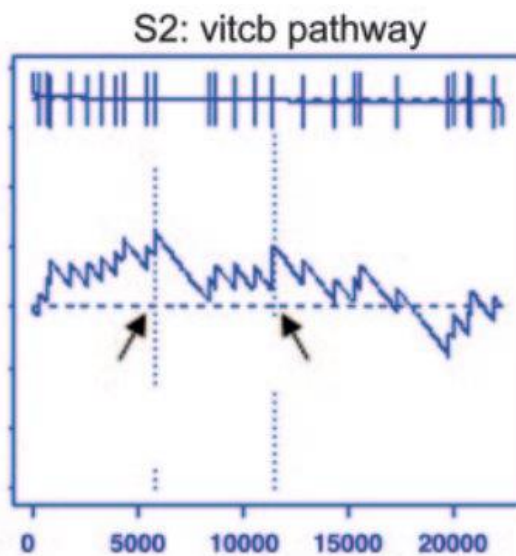
Gene symbol	Ranking
-------------	---------

Gene symbol	Ranking
-------------	---------

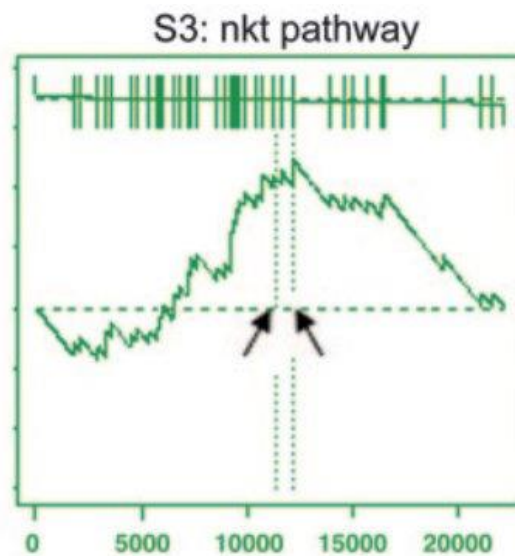
Gene symbol	Ranking
-------------	---------



Enriched towards the top



Randomly distributed



Not enriched towards the top but nonrandom

Three key elements of GSEA

Enrichment Score (ES)

Nominal p-value

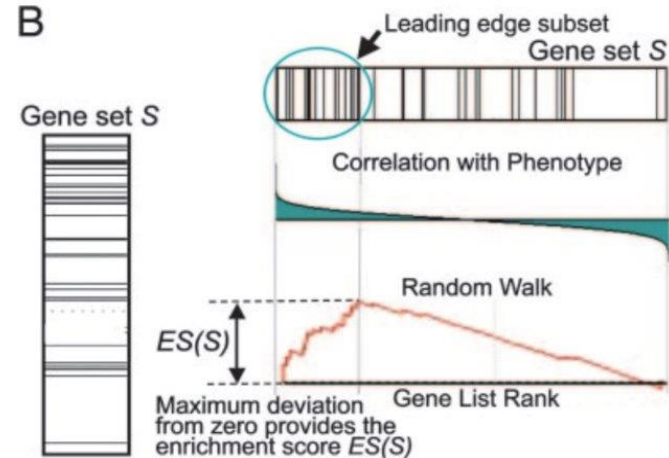
Multiple Hypothesis Correction

Enrichment Score (ES)

- It is the correlation of the gene set S with the phenotype, how much it is over-/under-represented in the ranked list L
- The score is the maximum deviation from 0, corresponding to Kolmogorov-Smirnov like statistic
- Calculated by walking down the list L , increasing the running-sum-statistic when a gene in S is encountered, decreasing when not

$$P_{\text{hit}}(S, i) = \sum_{\substack{g_j \in S \\ j \leq i}} \frac{|r_j|^p}{N_R}, \quad \text{where } N_R = \sum_{g_j \in S} |r_j|^p$$

$$P_{\text{miss}}(S, i) = \sum_{\substack{g_j \notin S \\ j \leq i}} \frac{1}{(N - N_H)}.$$



Nominal p-value

- Estimated by empirical phenotype-based permutation test
- Preserves gene-gene correlation and structure of gene expression data

- Permute phenotype label
- Construct null distribution
- Calculate p-value

Nominal p-value

actual data						
class label	tx	tx	tx	ctrl	ctrl	ctrl
sample name	sample 1	sample 2	sample 3	sample 4	sample 5	sample 6
permutation 1						
class label	tx	tx	ctrl	ctrl	ctrl	tx
sample name	sample 1	sample 2	sample 3	sample 4	sample 5	sample 6
permutation 2						
class label	tx	ctrl	ctrl	ctrl	tx	tx
sample name	sample 1	sample 2	sample 3	sample 4	sample 5	sample 6
...						
permutation n						
class label	ctrl	ctrl	ctrl	tx	tx	tx
sample name	sample 1	sample 2	sample 3	sample 4	sample 5	sample 6

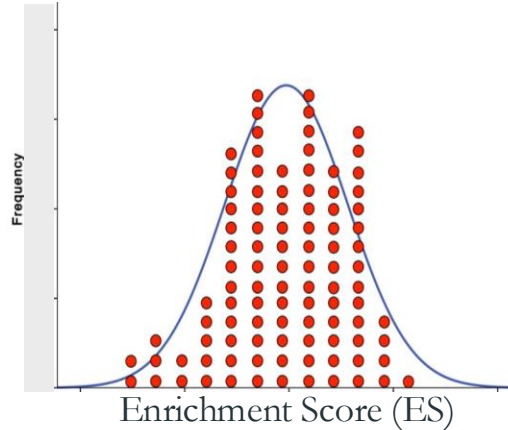
Permute phenotype label

actual data							
class label		tx	tx	tx	ctrl	ctrl	ctrl
sample name		sample 1	sample 2	sample 3	sample 4	sample 5	sample 6
permutation 1							
class label		tx	tx	ctrl	ctrl	ctrl	tx
sample name		sample 1	sample 2	sample 3	sample 4	sample 5	sample 6
permutation 2							
class label		tx	ctrl	ctrl	ctrl	tx	tx
sample name		sample 1	sample 2	sample 3	sample 4	sample 5	sample 6
...							
permutation n							
class label		ctrl	ctrl	ctrl	tx	tx	tx
sample name		sample 1	sample 2	sample 3	sample 4	sample 5	sample 6

Nominal p-value

actual data						
class label	tx	tx	tx	ctrl	ctrl	ctrl
sample name	sample 1	sample 2	sample 3	sample 4	sample 5	sample 6
permutation 1						
class label	tx	tx	ctrl	ctrl	ctrl	tx
sample name	sample 1	sample 2	sample 3	sample 4	sample 5	sample 6
permutation 2						
class label	tx	ctrl	ctrl	ctrl	tx	tx
sample name	sample 1	sample 2	sample 3	sample 4	sample 5	sample 6
...						
permutation n						
class label	ctrl	ctrl	ctrl	tx	tx	tx
sample name	sample 1	sample 2	sample 3	sample 4	sample 5	sample 6

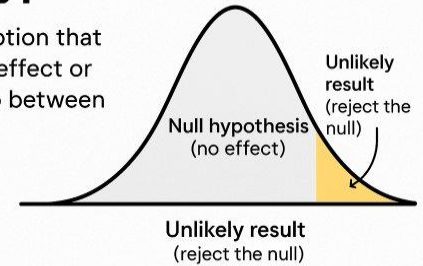
Permute phenotype label



Calculate ES for each permutation to form a null distribution

Null Hypothesis

The assumption that there is no effect or relationship between variables



Compare ES of observed data to null distribution to calculate p-value

Multiple Hypothesis Correction

- False Discovery Rate (FDR): The rate gene sets are considered significant at that particular Normalized Enrichment Score (NES) threshold are false positives

$$NES(S) = \frac{ES(S)}{\text{mean}_{\pi} [ES(S, \pi)]}$$

$$NES(S) = NES^*$$

$$FDR = \frac{\overbrace{\text{all gene sets} \times \text{all permutations}}^{\text{NULL}}}{\underbrace{\text{all gene sets}}_{\text{OBSERVED}}}$$

$$FDR(NES^*) = \frac{\text{fraction of null } NES(S, \pi) \geq NES^*}{\text{fraction of observed } NES(S) \geq NES^*}$$

$$FDR(NES^*) = \frac{\text{fraction of null } NES(S, \pi) \leq NES^*}{\text{fraction of observed } NES(S) \leq NES^*}$$

Leading edge subset

- "genes in the gene set S that appear in the ranked list L at, or before, the point where the running sum reaches its maximum deviation from zero"
- "core members of high scoring gene sets that contribute to the ES"
- Can be used to identify overlapping subset of genes within high scoring gene sets

Glossary

- Pathway: a set of genes that work together to carry out a biological function
- Pathway Analysis: method to interpret lists of genes in transcriptomic data
- Over-Representation Analysis (ORA): evaluates a *subset* of genes for over-/under-representation
- gProfiler, Metascape: web-based tools for ORA
- clusterProfiler: R package available on Bioconductor for ORA
- Gene Set Enrichment Analysis (GSEA): inputs *all* genes from the data and determines whether genes in a gene set are ranked towards the top/bottom
- Pathway topology: network graphs of pathways that account for directionality and pathway interactions

Glossary

- **Hypergeometric distribution**: sampling *without* replacement in a finite population
- **Binomial distribution**: sampling *with* replacement
- **Fisher's exact test**: checks association in *categorical* variables in a *small* population
- **Chi-Square test**: checks association in *categorical* variables in a *big* population
- **Kolmogorov-Smirnov test**: *non-parametric* test for *continuous* variables, no assumption of data, *sensitive* to and detects shifts in data shape, mean, variance
- **Wilcoxon rank sum**: *non-parametric* test for *continuous* variables, no assumption of data, robust to outliers and detects shifts in median
- **Maxmean**: statistic developed for GSA, calculates the average for positive and negative pathways separately then take the maxvalue

Glossary

- **Enrichment Score (ES)**: correlation to how much the gene set S with the phenotype is over-/under-represented in the ranked list, maximum deviation from 0
- **Normalized Enrichment Score (NES)**: ES of S /(divided by) mean ES of all permutation of gene set S
- **False Discovery Rate (FDR)**: rate of finding false positive result among the significant positive results, at FDR = 25% (0.25), $\frac{1}{4}$ result is a false positive
- **Leading edge subset**: common genes within high scoring gene sets

Resources (ctrl + click)

Pathway Analysis

- Ten years of pathway analysis: current approaches and outstanding challenges

Over Representation Analysis (ORA)

- Pathway enrichment analysis - simple explanation!

GSEA (FCS)

- Gene set enrichment analysis: A knowledge-based approach for interpreting genome-wide expression profiles
- Gene Set Enrichment Analysis (GSEA) – simply explained!
- Gene set enrichment analysis: performance evaluation and usage guidelines
- How to interpret GSEA results and plot - simple explanation of ES, NES, leading edge and more!

Thanks!

Do you have any questions?

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<https://github.com/juliana-ng>